

Results — v5 draft

Environmental Provisions in Chinese PTAs and Export Composition

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Standalone draft of Section 5, following the sequence: full panel → collapsed panel → comparison groups → sub-indices → dynamics, leave-one-out and the extensive margin. Five subsections, continuous prose, both margins carried together throughout. Citations render as small-caps placeholders (no bibliography in this file). Nothing in `paper_v3.tex` has been modified.

Specification recap. The estimating equation is

$$\ln x_{fpdt} = \beta_1 EP_{dt} \times g_p + \beta_2 EP_{dt} \times b_p + \gamma_1 TD_{dt} \times g_p + \gamma_2 TD_{dt} \times b_p + \theta_{fpd} + \theta_{fdt} + \theta_{pt} + \varepsilon_{fpdt}. \quad (1)$$

1 Results

1.1 The full panel

Table 1 reports the estimates of equation (1) on the full firm–product–destination–year panel. There are twelve coefficients of interest in it, and they are easier to read across than down. Columns (1)–(3) measure environmental content with the World Bank depth index and columns (4)–(6) with the TREND provision count; within each block the outcome is log export value, then log quantity, then log unit value. Panel A controls for non-environmental World Bank depth and Panel B for the independently coded DESTA index (DUR2014). The green and dirty interactions are estimated jointly, which is why they should be read together: the design asks whether an environmental chapter tilts a firm’s product mix, and a tilt has two ends.

Only one of those ends moves. On the green margin the coefficient is zero in every cell. Export value gives -0.0023 (s.e. 0.0039) with the World Bank index and -0.0001 (0.0010) with TREND; switching the depth control to DESTA changes the third decimal and nothing else. The joint F -test on all four composition interactions returns $p = 0.31$ (WB) and $p = 0.71$ (TREND) in Panel A, and $p = 0.18$ and $p = 0.26$ in Panel B. Quantity and unit value add nothing: across the eight green cells in columns (2), (3), (5) and (6) no coefficient comes close to significance, which rules out the one reading a value regression alone cannot dismiss — that a volume response and a price response are offsetting each other inside the value aggregate. Neither channel moves. If Chinese firms shifted toward environmental goods when an agreement with environmental content entered into force, the shift is not visible here at any margin.

The dirty coefficient does move, and it is worth being precise about how much. In Panel A it is -0.0044 on export value with an asymptotic p -value of 0.052, and -0.0009 with $p = 0.15$ under TREND. In Panel B the same two coefficients are -0.0056 ($p = 0.039$) and -0.0014 ($p = 0.048$). The sign is stable, the magnitude is stable, and the asymptotic verdict depends on which measure of non-environmental agreement depth sits in the regression. That dependence has a mechanical explanation.

Net of the fixed effects, environmental depth correlates 0.959 with World Bank non-environmental depth and 0.891 with DESTA depth. The World Bank control absorbs almost the same variation as the regressor of interest, so little independent variation is left and the standard error is wide; DESTA overlaps less, leaves more, and the standard error narrows. Nothing about the effect changes between the panels — only how much of it the design can see.

This is the point at which asymptotic p -values stop being informative. Standard errors clustered at the destination level are consistent as the number of clusters grows, and this design has 23 treated destinations sharing roughly 13 distinct agreement profiles, since the ten ASEAN partners enter under a single agreement. Under those conditions cluster-robust inference over-rejects, sometimes badly. The row labelled “Bootstrap p ” in every table of this paper reports the wild cluster bootstrap of CAMERON2008, implemented following RODMAN2019 with $B = 9,999$ replications: the model is re-estimated thousands of times with the residuals of each destination flipped in sign at random, and the observed statistic is placed in the distribution this generates. It is the reference inference throughout, and the asymptotic figures are kept alongside it only so the gap between the two remains visible.

The gap is large exactly where it matters. On the green margin the bootstrap changes nothing: p moves from 0.57 to 0.69 (WB) and from 0.92 to 0.93 (TREND), and the null is a null either way. On the dirty margin under the World Bank control it changes the verdict, taking p from 0.052 to 0.19 and from 0.15 to 0.18. Under DESTA it does not: 0.039 becomes 0.049 and 0.048 becomes 0.069. So the dirty coefficient survives robust inference under one depth control and not under the other, and it does so on both indices at once, which at least means the two codings are telling the same story. The one place where it is strongest is quantity: under DESTA the World Bank interaction is -0.0070 on log quantity with a bootstrap p of 0.035, larger and better determined than on value, while the unit value coefficient is $+0.0004$ and nowhere near significance. Whatever this is, it is a shipment volume, not a price.

The sign, at least, is where the recent literature on China would put it. YUE2024 find that environmental provisions in Chinese PTAs inhibit exports of energy-consuming products, and ZHUSUN2026, working with customs data that overlap with those used here, report that environmental provisions reduce dirty exports and raise green ones. On dirty the present estimates agree in sign and are considerably smaller; on green they do not agree at all. The difference is not mysterious. Their identification comes from variation across firms and destinations, which is precisely the variation that θ_{fdt} removes here. Section 1.2 measures how much of the association that channel accounts for. ZHUSUN2025 report that environmental provisions are associated with quality upgrading in Chinese firms’ exports; unit value is a coarse proxy for quality and column (3) is null, but the outcome and the sample differ enough that this should not be read as a test of their result.

Before going further it is worth asking how much of Table 1 is the specification rather than the data. Table 2 re-estimates the same interactions with progressively different sets of absorbed effects. The pattern is instructive. In columns (1)–(4) the coefficients are frequently positive and occasionally significant: the World Bank dirty interaction reaches $+0.0064$ ($p = 0.019$) and the TREND green interaction $+0.0021$ ($p = 0.021$). None of those four specifications contains a fixed effect with both a destination and a time dimension. Chinese exports to a partner grow around the entry into force of an agreement for reasons that have nothing to do with its environmental chapter, and without a destination–time absorber that growth loads onto whichever interaction is available. Column (5) adds θ_{fdt} and the coefficients collapse to the values of Table 1. The saturated specification is not one choice among five; it is the only one in the sequence that removes the obvious confounder, and the others are shown to make clear what it removes.

Table 1: Full panel — triple-difference estimates

	WB EP depth (EP_{dt}^{WB})			TREND EP count (EP_{dt}^{TR})		
	(1) $\ln x_{fpdt}^{val}$	(2) $\ln x_{fpdt}^{qua}$	(3) $\ln x_{fpdt}^{uv}$	(4) $\ln x_{fpdt}^{val}$	(5) $\ln x_{fpdt}^{qua}$	(6) $\ln x_{fpdt}^{uv}$
<i>Panel A — Non-environmental depth control (TD_{dt}, World Bank)</i>						
$EP_{dt} \times g_p$	−0.0023 (0.0039)	−0.0013 (0.0041)	−0.0004 (0.0013)	−0.0001 (0.0010)	−0.00003 (0.0011)	−0.0001 (0.0003)
Asymptotic p	[0.566]	[0.750]	[0.748]	[0.924]	[0.974]	[0.857]
Bootstrap p	[0.686]	[0.851]	[0.740]	[0.931]	[0.978]	[0.860]
$EP_{dt} \times b_p$	−0.0044* (0.0022)	−0.0048* (0.0029)	0.0001 (0.0010)	−0.0009 (0.0006)	−0.0006 (0.0007)	−0.0001 (0.0004)
Asymptotic p	[0.052]	[0.096]	[0.903]	[0.153]	[0.409]	[0.697]
Bootstrap p	[0.185]	[0.391]	[0.937]	[0.177]	[0.443]	[0.734]
θ_{fpd}	✓	✓	✓	✓	✓	✓
$\theta_{f dt}$	✓	✓	✓	✓	✓	✓
θ_{pt}	✓	✓	✓	✓	✓	✓
Observations	21,519,511	21,313,047	21,313,037	21,519,511	21,313,047	21,313,037
Clusters (d)	225	225	225	225	225	225
R^2	0.873	0.913	0.963	0.873	0.913	0.963
<i>Panel B — DESTA depth control</i>						
$EP_{dt} \times g_p$	−0.0019 (0.0037)	−0.0015 (0.0034)	−0.0004 (0.0019)	−0.0001 (0.0010)	−0.0001 (0.0010)	−0.0001 (0.0005)
Asymptotic p	[0.613]	[0.660]	[0.819]	[0.904]	[0.917]	[0.776]
Bootstrap p	[0.673]	[0.755]	[0.843]	[0.920]	[0.933]	[0.852]
$EP_{dt} \times b_p$	−0.0056** (0.0027)	−0.0070** (0.0030)	0.0004 (0.0011)	−0.0014** (0.0007)	−0.0014* (0.0008)	−0.00004 (0.0003)
Asymptotic p	[0.039]	[0.020]	[0.676]	[0.048]	[0.080]	[0.903]
Bootstrap p	[0.049]	[0.035]	[0.765]	[0.069]	[0.100]	[0.915]
θ_{fpd}	✓	✓	✓	✓	✓	✓
$\theta_{f dt}$	✓	✓	✓	✓	✓	✓
θ_{pt}	✓	✓	✓	✓	✓	✓
Observations	21,517,666	21,311,206	21,311,196	21,517,666	21,311,206	21,311,196
Clusters (d)	225	225	225	225	225	225
R^2	0.873	0.913	0.963	0.873	0.913	0.963

Notes: Full firm-level panel, excluding Hong Kong and Macao. Columns (1) and (4): log export value ($\ln x_{fpdt}^{val}$). Columns (2) and (5): log export quantity ($\ln x_{fpdt}^{qua}$). Columns (3) and (6): log unit value ($\ln x_{fpdt}^{uv} = \ln x_{fpdt}^{val} - \ln x_{fpdt}^{qua}$). Quantity and unit-value regressions are estimated on the subset of observations with non-missing quantity data. All specifications include $TD_{dt} \times g_p$ and $TD_{dt} \times b_p$ controls (not shown). Panel A uses non-environmental WB depth (TD_{dt}); Panel B uses DESTA depth. Standard errors clustered at the destination level in parentheses. Wild cluster bootstrap ($B = 9,999$) p -values in brackets. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ (asymptotic).

Source: Chinese Customs Office; World Bank DTA Database; TREND Database.

Table 2: Fixed-effects structure and the composition interactions (full panel, log export value)

	(1)	(2)	(3)	(4)	(5)
$EP_{dt}^{WB} \times g_p$	0.0018 (0.0050)	0.0012 (0.0027)	0.0060 (0.0050)	0.0050 (0.0037)	-0.0023 (0.0039)
$EP_{dt}^{WB} \times b_p$	-0.0037 (0.0150)	-0.0015 (0.0027)	0.0058* (0.0031)	0.0064** (0.0027)	-0.0044* (0.0022)
$EP_{dt}^{TR} \times g_p$	-0.0013 (0.0010)	0.0010 (0.0009)	0.0021** (0.0009)	0.0018** (0.0008)	-0.0001 (0.0010)
$EP_{dt}^{TR} \times b_p$	-0.0049*** (0.0015)	0.00003 (0.0007)	0.0002 (0.0011)	0.0009 (0.0008)	-0.0009 (0.0006)
θ_{fpd}	✓		✓	✓	✓
θ_{fpt}		✓	✓		
θ_{pd}		✓			
θ_{pt}				✓	✓
θ_t	✓				
θ_{fdt}					✓
Absorbs destination×time?	No	No	No	No	Yes
Observations	29,477,365	35,607,036	22,927,402	29,473,145	21,519,511
Clusters (d)	231	232	231	231	225

Notes: Each column is a separate estimation of the composition interactions on the full firm-level panel with a different set of absorbed fixed effects; the dependent variable is log export value throughout. Column (5) is the reference specification of Table 1, Panel A, column (1), reproduced here so the sequence terminates at it. Columns (1)–(4) differ in observation count because a less saturated fixed-effects structure removes fewer singleton cells. Hong Kong and Macao excluded throughout. All specifications include $TD_{dt} \times g_p$ and $TD_{dt} \times b_p$ controls (not shown). Standard errors clustered at the destination level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ (asymptotic).

Columns (1)–(4) contain no fixed effect with both a destination and a time dimension, so destination-specific shocks that move with the timing of an agreement are not absorbed. Column (5) adds θ_{fdt} . Quantity and unit-value outcomes, and the versions including Hong Kong and Macao, are reported in Appendix Tables 1.5 and 1.5.

Source: Chinese Customs Office; World Bank DTA Database; TREND Database.

1.2 The collapsed panel

Table 3 repeats the exercise on HS6–destination–year cells weighted by cell size. Three things change at once and it is worth naming them before reading the numbers. The unit of observation is a product–destination–year cell rather than a firm–product–destination–year shipment; the dependent variable is therefore a cell mean of log exports rather than an individual flow; and the fixed effects become $\theta_{pd} + \theta_{dt} + \theta_{pt}$, since firm identifiers no longer exist in the collapsed data. The last of these is the substantive change. The firm dimension of the absorbers is gone, so a shift in *which* firms ship dirty goods to a destination now enters the coefficient, where in Table 1 it was absorbed. The collapsed panel is not a robustness check on the full panel; it estimates a wider object, and the difference between the two is informative in its own right.

On green the two panels agree, as they have agreed everywhere so far: -0.0046 (bootstrap $p = 0.65$) with the World Bank index, $+0.0018$ ($p = 0.39$) with TREND, and again nothing on quantity or unit value. On dirty they do not. The World Bank interaction is -0.0119 against -0.0044 in the full panel, a factor of 2.7, with an asymptotic p -value below 0.001. Since the two specifications differ by exactly the between-firm channel, the gap measures it: $(0.0119 - 0.0044)/0.0119 = 0.63$, so roughly three-fifths of the aggregate dirty association is a change in the composition of exporters rather than reallocation inside a given firm. This is the quantity that reconciles the present null with ZHUSUN2026. Their design keeps that channel; equation (1) removes it. Neither is wrong, and the two results are not in conflict once the estimands are stated.

The collapsed panel is also where the design can be attacked with a third test. The wild cluster bootstrap already raises the dirty p -value from below 0.001 to 0.072, which is a large correction, but the bootstrap still takes the assignment of treatment as given and only resamples the residuals. The permutation test asks a blunter question. The environmental profile of each agreement — its depth and the years in which it takes effect — is detached from its destination and reassigned at random among the 23 treated destinations; the model is re-estimated; and the exercise is repeated a thousand times. The reported p -value is the share of these random reassignments that produce a coefficient at least as large in absolute value as the observed one. It is expensive, which is why it is run on export value only, and it is discrete, because thirteen distinct profiles do not generate a smooth distribution. But it answers the question a reader of this table actually has: would this number have appeared if we had simply relabelled which of these destinations signed environmental chapters?

For the dirty coefficient the answer is that it would, 27.8% of the time. For green it is 59.7%. Under the DESTA control the two figures are 14.6% and 48.0%. This is the strongest evidence in the paper against a causal reading of the dirty coefficient, and it is strong precisely because it is indifferent to the choice that Section 1.1 could not resolve: the permutation distribution is built by shuffling treatment, not by changing the control, so it gives the same verdict under both depth measures. The dirty association is real in the sense that it is in the data; it is not distinguishable from what random relabelling of these same destinations produces.

Table 3: Collapsed panel — triple-difference estimates

	WB EP depth (EP_{dt}^{WB})			TREND EP count (EP_{dt}^{TR})		
	(1) $\ln x_{fpdt}^{val}$	(2) $\ln x_{fpdt}^{qua}$	(3) $\ln x_{fpdt}^{uv}$	(4) $\ln x_{fpdt}^{val}$	(5) $\ln x_{fpdt}^{qua}$	(6) $\ln x_{fpdt}^{uv}$
<i>Panel A — Non-environmental depth control (TD_{dt}, World Bank)</i>						
$EP_{dt} \times g_p$	−0.0046 (0.0069)	−0.0055 (0.0069)	0.0005 (0.0028)	0.0018 (0.0018)	0.0019 (0.0020)	−0.0001 (0.0007)
Asymptotic p	[0.508]	[0.426]	[0.854]	[0.316]	[0.360]	[0.868]
Bootstrap p	[0.649]	[0.525]	[0.851]	[0.390]	[0.433]	[0.867]
Permutation p	[0.597]			[0.160]		
$EP_{dt} \times b_p$	−0.0119*** (0.0029)	−0.0115* (0.0067)	0.0006 (0.0042)	0.0004 (0.0016)	−0.0004 (0.0019)	0.0009 (0.0005)
Asymptotic p	[0.000]	[0.087]	[0.880]	[0.825]	[0.837]	[0.076]
Bootstrap p	[0.072]	[0.250]	[0.947]	[0.857]	[0.886]	[0.169]
Permutation p	[0.278]			[0.817]		
θ_{pd}	✓	✓	✓	✓	✓	✓
θ_{dt}	✓	✓	✓	✓	✓	✓
θ_{pt}	✓	✓	✓	✓	✓	✓
Observations	3,681,023	3,672,709	3,672,709	3,681,023	3,672,709	3,672,709
Clusters (d)	228	228	228	228	228	228
R^2	0.843	0.919	0.966	0.843	0.919	0.966
<i>Panel B — DESTA depth control</i>						
$EP_{dt} \times g_p$	−0.0043 (0.0043)	−0.0084* (0.0044)	0.0038 (0.0042)	0.0016 (0.0020)	0.0010 (0.0022)	0.0006 (0.0008)
Asymptotic p	[0.320]	[0.058]	[0.368]	[0.428]	[0.649]	[0.495]
Bootstrap p	[0.546]	[0.268]	[0.446]	[0.508]	[0.712]	[0.542]
Permutation p	[0.480]			[0.316]		
$EP_{dt} \times b_p$	−0.0113*** (0.0022)	−0.0097** (0.0045)	−0.0012 (0.0029)	−0.0002 (0.0014)	−0.0008 (0.0014)	0.0007 (0.0006)
Asymptotic p	[0.000]	[0.033]	[0.689]	[0.858]	[0.603]	[0.252]
Bootstrap p	[0.047]	[0.311]	[0.757]	[0.905]	[0.876]	[0.332]
Permutation p	[0.146]			[0.904]		
θ_{pd}	✓	✓	✓	✓	✓	✓
θ_{dt}	✓	✓	✓	✓	✓	✓
θ_{pt}	✓	✓	✓	✓	✓	✓
Observations	3,677,333	3,669,025	3,669,025	3,677,333	3,669,025	3,669,025
Clusters (d)	228	228	228	228	228	228
R^2	0.843	0.919	0.966	0.843	0.919	0.966

Notes: Collapsed panel (product×destination×period cells, analytic weights), excluding Hong Kong and Macao. Columns (1) and (4): log export value ($\ln x_{fpdt}^{val}$). Columns (2) and (5): log export quantity ($\ln x_{fpdt}^{qua}$). Columns (3) and (6): log unit value ($\ln x_{fpdt}^{uv}$). All specifications include $TD_{dt} \times g_p$ and $TD_{dt} \times b_p$ controls (not shown). Panel A uses non-environmental WB depth (TD_{dt}); Panel B uses DESTA depth. Standard errors clustered at the destination level in parentheses. Wild cluster bootstrap ($B = 9,999$) p -values in brackets. Permutation p -values (treated-only design, $R = 1,000$) reported for export value only; columns (2)–(3) and (5)–(6) not permuted. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ (asymptotic).

Source: Chinese Customs Office; World Bank DTA Database; TREND Database.

1.3 Changing the comparison group

Everything so far compares green and dirty products against neutral products, and agreement destinations against non-agreement destinations. Both comparisons can be questioned, and the next three tables replace them one at a time.

Neutral products may be poor counterfactuals for green ones if the two sit in different regions of the product space and are exposed to different shocks. Table 4 therefore restricts the sample to HS6 lines with a comparable match at the four-digit level, so that a solar panel is compared with its neighbours inside the same heading rather than with the entire neutral universe. The green coefficient becomes -0.0009 with the World Bank index and $+0.0005$ with TREND, both with bootstrap p -values above 0.72. The dirty coefficient does something more interesting: it changes sign, to $+0.0066$ and $+0.0033$, and remains insignificant. Narrowing the product comparison does not shrink the dirty estimate towards zero so much as scramble it, which is what one expects of a coefficient that is not identifying a stable object.

Destinations raise the symmetric concern. China does not sign agreements at random, and if partners are on different growth paths than non-partners the comparison is contaminated. Table 5 matches destinations by coarsened exact matching on log GDP per capita, growth and the MFN tariff in 2000, which leaves 57 destination clusters that are comparable on pre-treatment observables. Both margins survive intact: green is -0.0023 (WB) and -0.0006 (TREND) with bootstrap p -values of 0.61, and dirty is -0.0041 and -0.0011 with bootstrap p -values of 0.25 and 0.20. Under the DESTA control the dirty coefficients are -0.0043 and -0.0013 , with bootstrap p -values of 0.12 and 0.13 — weaker than in the full sample, which is the opposite of what a genuine effect concentrated among comparable partners would produce.

The third table removes the untreated comparison group altogether. In Table 6 the sample is restricted to the 23 PTA partners, so that identification comes from variation in environmental depth among destinations that all have an agreement in force: a shallow environmental chapter is the counterfactual for a deep one. This is the most demanding of the three designs and the one with the fewest clusters. Every coefficient is small and none is significant — green -0.0022 and -0.0004 , dirty -0.0034 and -0.0006 under the World Bank control, with bootstrap p -values between 0.50 and 0.75.

Read together, the three tables say two different things about the two margins. The green null does not depend on the comparison group: across the six estimates in this section it stays between -0.0023 and $+0.0005$ and its bootstrap p -value never falls below 0.58. An artifact of the control group would move when the control group is replaced three times over; this does not move. The dirty coefficient, by contrast, survives none of the three replacements. It keeps its sign under matching and among partners, loses it under the within-heading product comparison, and is insignificant in all of them. Combined with the permutation result of Section 1.2, the reading that fits the evidence is that the dirty coefficient is a feature of the specification in which it appears rather than a composition effect that would show up wherever one looked.

Table 4: C-prod-HS4 subsample — triple-difference estimates

	WB EP depth (EP_{dt}^{WB})			TREND EP count (EP_{dt}^{TR})		
	(1) $\ln x_{fpdt}^{val}$	(2) $\ln x_{fpdt}^{qua}$	(3) $\ln x_{fpdt}^{uv}$	(4) $\ln x_{fpdt}^{val}$	(5) $\ln x_{fpdt}^{qua}$	(6) $\ln x_{fpdt}^{uv}$
<i>Panel A — Non-environmental depth control (TD_{dt}, World Bank)</i>						
$EP_{dt} \times g_p$	−0.0009 (0.0051)	−0.0019 (0.0043)	0.0005 (0.0007)	0.0005 (0.0011)	0.0001 (0.0010)	0.0002 (0.0004)
Asymptotic p	[0.858]	[0.665]	[0.471]	[0.680]	[0.906]	[0.689]
Bootstrap p	[0.860]	[0.662]	[0.421]	[0.730]	[0.916]	[0.713]
$EP_{dt} \times b_p$	0.0066 (0.0090)	0.0068 (0.0111)	−0.0017 (0.0070)	0.0024 (0.0066)	0.0033 (0.0059)	−0.0004 (0.0024)
Asymptotic p	[0.469]	[0.541]	[0.807]	[0.714]	[0.581]	[0.860]
Bootstrap p	[0.435]	[0.562]	[0.857]	[0.734]	[0.591]	[0.866]
θ_{fpd}	✓	✓	✓	✓	✓	✓
θ_{fdt}	✓	✓	✓	✓	✓	✓
θ_{pt}	✓	✓	✓	✓	✓	✓
Observations	3,772,855	3,745,788	3,745,788	3,772,855	3,745,788	3,745,788
Clusters (d)	215	215	215	215	215	215
Adj. R^2	0.737	0.835	0.923	0.737	0.835	0.923
<i>Panel B — DESTA depth control</i>						
$EP_{dt} \times g_p$	−0.0009 (0.0051)	−0.0032 (0.0042)	0.0012 (0.0012)	0.0004 (0.0013)	−0.0003 (0.0011)	0.0003 (0.0004)
Asymptotic p	[0.867]	[0.449]	[0.338]	[0.779]	[0.790]	[0.484]
Bootstrap p	[0.855]	[0.461]	[0.368]	[0.817]	[0.806]	[0.546]
$EP_{dt} \times b_p$	0.0033 (0.0168)	−0.0059 (0.0182)	0.0076 (0.0098)	0.0015 (0.0040)	−0.0006 (0.0032)	0.0022 (0.0025)
Asymptotic p	[0.846]	[0.747]	[0.440]	[0.719]	[0.854]	[0.383]
Bootstrap p	[0.850]	[0.758]	[0.554]	[0.735]	[0.852]	[0.454]
θ_{fpd}	✓	✓	✓	✓	✓	✓
θ_{fdt}	✓	✓	✓	✓	✓	✓
θ_{pt}	✓	✓	✓	✓	✓	✓
Observations	3,772,321	3,745,254	3,745,254	3,772,321	3,745,254	3,745,254
Clusters (d)	214	214	214	214	214	214
Adj. R^2	0.737	0.835	0.923	0.737	0.835	0.923

Notes: C-prod-HS4 subsample: restricted to HS6 products with a comparable HS4-level match across classification revisions. Full firm-level panel, excluding Hong Kong and Macao. All specifications include $TD_{dt} \times g_p$ and $TD_{dt} \times b_p$ controls (not shown). Panel A uses non-environmental WB depth (TD_{dt}); Panel B uses DESTA depth. Standard errors clustered at the destination level in parentheses. Wild cluster bootstrap ($B = 9,999$) p -values in brackets. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ (asymptotic).

Source: Chinese Customs Office; World Bank DTA Database; TREND Database.

Table 5: CEM destinations — triple-difference estimates

	WB EP depth (EP_{dt}^{WB})			TREND EP count (EP_{dt}^{TR})		
	(1) $\ln x_{fpdt}^{val}$	(2) $\ln x_{fpdt}^{qua}$	(3) $\ln x_{fpdt}^{uv}$	(4) $\ln x_{fpdt}^{val}$	(5) $\ln x_{fpdt}^{qua}$	(6) $\ln x_{fpdt}^{uv}$
<i>Panel A — Non-environmental depth control (TD_{dt}, World Bank)</i>						
$EP_{dt} \times g_p$	−0.0023 (0.0037)	−0.0014 (0.0041)	−0.0003 (0.0016)	−0.0006 (0.0011)	−0.0002 (0.0012)	−0.0004 (0.0003)
Asymptotic p	[0.543]	[0.734]	[0.837]	[0.576]	[0.857]	[0.201]
Bootstrap p	[0.610]	[0.824]	[0.887]	[0.609]	[0.878]	[0.241]
$EP_{dt} \times b_p$	−0.0041* (0.0024)	−0.0042 (0.0034)	−0.0002 (0.0013)	−0.0011 (0.0007)	−0.0006 (0.0010)	−0.0004 (0.0003)
Asymptotic p	[0.098]	[0.214]	[0.868]	[0.140]	[0.541]	[0.308]
Bootstrap p	[0.251]	[0.544]	[0.928]	[0.204]	[0.610]	[0.403]
θ_{fpd}	✓	✓	✓	✓	✓	✓
θ_{fdt}	✓	✓	✓	✓	✓	✓
θ_{pt}	✓	✓	✓	✓	✓	✓
Observations	13,992,396	13,858,213	13,858,206	13,992,396	13,858,213	13,858,206
Clusters (d)	57	57	57	57	57	57
Adj. R^2	0.742	0.815	0.921	0.742	0.815	0.921
<i>Panel B — DESTA depth control</i>						
$EP_{dt} \times g_p$	−0.0016 (0.0034)	−0.0016 (0.0035)	0.0000 (0.0019)	−0.0005 (0.0011)	−0.0003 (0.0011)	−0.0003 (0.0005)
Asymptotic p	[0.648]	[0.644]	[0.980]	[0.655]	[0.783]	[0.567]
Bootstrap p	[0.676]	[0.737]	[0.978]	[0.676]	[0.815]	[0.683]
$EP_{dt} \times b_p$	−0.0043* (0.0024)	−0.0055* (0.0030)	0.0001 (0.0013)	−0.0013* (0.0007)	−0.0011 (0.0009)	−0.0002 (0.0003)
Asymptotic p	[0.081]	[0.071]	[0.938]	[0.091]	[0.194]	[0.478]
Bootstrap p	[0.121]	[0.180]	[0.981]	[0.125]	[0.255]	[0.607]
θ_{fpd}	✓	✓	✓	✓	✓	✓
θ_{fdt}	✓	✓	✓	✓	✓	✓
θ_{pt}	✓	✓	✓	✓	✓	✓
Observations	13,992,396	13,858,213	13,858,206	13,992,396	13,858,213	13,858,206
Clusters (d)	57	57	57	57	57	57
Adj. R^2	0.742	0.815	0.921	0.742	0.815	0.921

Notes: CEM-matched destinations subsample. Full firm-level panel, excluding Hong Kong and Macao. Columns (1) and (4): log export value ($\ln x_{fpdt}^{val}$). Columns (2) and (5): log export quantity ($\ln x_{fpdt}^{qua}$). Columns (3) and (6): log unit value ($\ln x_{fpdt}^{uv}$). Quantity and unit-value regressions estimated on the subset with non-missing quantity data. All specifications include $TD_{dt} \times g_p$ and $TD_{dt} \times b_p$ controls (not shown). Panel A uses non-environmental WB depth (TD_{dt}); Panel B uses DESTA depth. Standard errors clustered at the destination level in parentheses. Wild cluster bootstrap ($B = 9,999$) p -values in brackets. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ (asymptotic).

Source: Chinese Customs Office; World Bank DTA Database; TREND Database.

Table 6: Deep vs. shallow agreements — triple-difference estimates

	WB EP depth (EP_{dt}^{WB})			TREND EP count (EP_{dt}^{TR})		
	(1) $\ln x_{fpdt}^{val}$	(2) $\ln x_{fpdt}^{qua}$	(3) $\ln x_{fpdt}^{uv}$	(4) $\ln x_{fpdt}^{val}$	(5) $\ln x_{fpdt}^{qua}$	(6) $\ln x_{fpdt}^{uv}$
<i>Panel A — Non-environmental depth control (TD_{dt}, World Bank)</i>						
$EP_{dt} \times g_p$	−0.0022 (0.0035)	−0.0013 (0.0036)	−0.0004 (0.0011)	−0.0004 (0.0010)	−0.0001 (0.0010)	−0.0002 (0.0002)
Asymptotic p	[0.534]	[0.712]	[0.748]	[0.721]	[0.903]	[0.392]
Bootstrap p	[0.585]	[0.817]	[0.708]	[0.751]	[0.919]	[0.305]
$EP_{dt} \times b_p$	−0.0034 (0.0029)	−0.0034 (0.0034)	−0.0005 (0.0008)	−0.0006 (0.0010)	−0.0004 (0.0012)	−0.0003 (0.0003)
Asymptotic p	[0.246]	[0.330]	[0.572]	[0.549]	[0.712]	[0.389]
Bootstrap p	[0.499]	[0.601]	[0.771]	[0.739]	[0.785]	[0.379]
θ_{fpd}	✓	✓	✓	✓	✓	✓
θ_{fdt}	✓	✓	✓	✓	✓	✓
θ_{pt}	✓	✓	✓	✓	✓	✓
Observations	5,262,293	5,201,289	5,201,282	5,262,293	5,201,289	5,201,282
Clusters (d)	23	23	23	23	23	23
Adj. R^2	0.751	0.844	0.932	0.751	0.844	0.932
<i>Panel B — DESTA depth control</i>						
$EP_{dt} \times g_p$	−0.0016 (0.0032)	−0.0015 (0.0031)	−0.0002 (0.0018)	−0.0003 (0.0010)	−0.0002 (0.0009)	−0.0002 (0.0004)
Asymptotic p	[0.619]	[0.633]	[0.915]	[0.776]	[0.832]	[0.624]
Bootstrap p	[0.616]	[0.727]	[0.910]	[0.784]	[0.857]	[0.715]
$EP_{dt} \times b_p$	−0.0042 (0.0025)	−0.0043 (0.0028)	−0.0007 (0.0009)	−0.0009 (0.0008)	−0.0008 (0.0009)	−0.0003 (0.0003)
Asymptotic p	[0.113]	[0.141]	[0.431]	[0.253]	[0.401]	[0.294]
Bootstrap p	[0.173]	[0.294]	[0.755]	[0.386]	[0.492]	[0.336]
θ_{fpd}	✓	✓	✓	✓	✓	✓
θ_{fdt}	✓	✓	✓	✓	✓	✓
θ_{pt}	✓	✓	✓	✓	✓	✓
Observations	5,260,451	5,199,451	5,199,444	5,260,451	5,199,451	5,199,444
Clusters (d)	23	23	23	23	23	23
Adj. R^2	0.751	0.844	0.932	0.751	0.844	0.932

Notes: Deep vs. shallow agreements subsample: restricted to destinations with at least one deep PTA in force. Full firm-level panel, excluding Hong Kong and Macao. Columns (1) and (4): log export value ($\ln x_{fpdt}^{val}$). Columns (2) and (5): log export quantity ($\ln x_{fpdt}^{qua}$). Columns (3) and (6): log unit value ($\ln x_{fpdt}^{uv}$). Quantity and unit-value regressions estimated on the subset with non-missing quantity data. All specifications include $TD_{dt} \times g_p$ and $TD_{dt} \times b_p$ controls (not shown). Panel A uses non-environmental WB depth (TD_{dt}); Panel B uses DESTA depth. Standard errors clustered at the destination level in parentheses. Wild cluster bootstrap ($B = 9,999$) p -values in brackets. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ (asymptotic).

Source: Chinese Customs Office; World Bank DTA Database; TREND Database.

1.4 Does the content of the agreement matter?

One reading of the null so far is that it is an artifact of aggregation. The indices used to this point add up every environmental provision in an agreement regardless of what it does, and most of what they add up is cooperation language, institutional machinery and general environmental statements — text with no channel through which it could move a product mix. Only a small minority of provisions carry a trade mechanism: tariff reductions on environmental goods, non-regression clauses that prevent a partner from relaxing standards to attract trade, binding market-access commitments. If the few clauses that could bite are diluted by the many that cannot, a null on the aggregate index would say nothing about whether environmental content matters.

The dilution is severe. The World Bank index checks 150 provision slots across the treated destinations, and 8 of them — 5.3% — fall in sub-indices with a trade mechanism. TREND, which codes at finer granularity, gives 4 of 437, or 0.9%. At the level of the agreement, a binding environmental obligation appears in exactly one of the fourteen Chinese PTAs in the database, the 2015 agreement with Korea. This is worth stating plainly because it bounds what any heterogeneity analysis on this sample can deliver: the two World Bank sub-indices with a trade mechanism are non-zero in only three destination-years and always in the same proportion, which makes them perfectly collinear and impossible to separate. BRANDI2020 report a maximum variance inflation factor of 4.6 across provision types on a panel of 680 agreements; with thirteen distinct profiles there is no equivalent here.

What can still be done is to replace the aggregate index with one sub-index at a time and re-estimate. Table 7 does this on the full panel for seven sub-indices, two from the World Bank coding and five from TREND. The coefficients are not partial effects holding other clause types fixed; each absorbs whatever correlated variation moves with it, so the table is a set of separate comparisons rather than a decomposition. It should be read for pattern, not for magnitude.

The pattern is the same one as before. On the green margin nothing registers: the two trade-mechanism sub-indices give +0.0085 (World Bank Green Liberalization, $p = 0.86$) and +0.0158 (TREND Green Market Access, $p = 0.38$), and the remaining five are equally silent. On the dirty margin two sub-indices are marginally negative under asymptotic inference — enforcement and dispute settlement attached to the environmental chapter, at -0.0243 ($p = 0.040$) in the World Bank coding and -0.0107 ($p = 0.081$) in TREND — and this is the same asymptotics-only signal documented in Section 1.1, on a smaller and more collinear regressor. The wild cluster bootstrap was not run on the full-panel sub-indices, so these two figures are asymptotic and should be discounted accordingly.¹

The conclusion is a negative one and it is the honest one: this sample cannot tell whether the null reflects the absence of an effect or the absence of provisions capable of producing one. What it can establish is that the two are observationally equivalent here, and that a null on the aggregate index is exactly what BRANDI2020's own heterogeneity would predict for a set of agreements composed almost entirely of cooperation clauses.

¹On the collapsed panel, where the bootstrap is available for the sub-indices, the only coefficient that survives it is the TREND regulatory-space index, at +0.024 on green ($p = 0.045$) and +0.023 on dirty ($p = 0.021$). Both margins move together and by the same amount, so the green-dirty differential is +0.0017 with $p = 0.80$: this is not a shift in composition toward cleaner goods but a proportional rise in both categories against the neutral baseline. The regulatory-space index also accounts for 71.5% of the TREND count and correlates 0.90 with total depth, which makes it a poor placebo rather than a clean channel. Appendix Table 1.5 reports the full set.

Table 7: Full panel — sub-index decomposition

	WB sub-indices		TREND sub-indices				
	(1) Green Lib.	(2) Enf. DSM	(3) Green Mkt.	(4) Enf. DSM	(5) Hard	(6) Soft	(7) Reg. Space
<i>Panel A — Non-environmental depth control (TD_{dt}, World Bank)</i>							
$EP_{\text{sub}} \times g_p$	0.0085 (0.0467)	−0.0028 (0.0208)	0.0158 (0.0178)	0.0020 (0.0083)	0.0007 (0.0023)	−0.0037 (0.0085)	0.0049 (0.0095)
$EP_{\text{sub}} \times b_p$	−0.0019 (0.0335)	−0.0243** (0.0117)	−0.0088 (0.0153)	−0.0107* (0.0061)	−0.0019 (0.0016)	−0.0047 (0.0070)	0.0072 (0.0075)
$\theta_{f_{pd}}$	✓	✓	✓	✓	✓	✓	✓
$\theta_{f_{dt}}$	✓	✓	✓	✓	✓	✓	✓
θ_{pt}	✓	✓	✓	✓	✓	✓	✓
Observations	21,519,511	21,519,511	21,519,511	21,519,511	21,519,511	21,519,511	21,519,511
Clusters (d)	225	225	225	225	225	225	225
Adj. R^2	0.744	0.744	0.744	0.744	0.744	0.744	0.744
<i>Panel B — DESTA depth control</i>							
$EP_{\text{sub}} \times g_p$	0.0073 (0.0432)	−0.0029 (0.0168)	0.0141 (0.0162)	0.0016 (0.0087)	0.0005 (0.0023)	−0.0038 (0.0077)	0.0029 (0.0068)
$EP_{\text{sub}} \times b_p$	−0.0171 (0.0316)	−0.0308** (0.0150)	−0.0155 (0.0143)	−0.0133* (0.0068)	−0.0029 (0.0018)	−0.0080* (0.0043)	−0.0037 (0.0060)
$\theta_{f_{pd}}$	✓	✓	✓	✓	✓	✓	✓
$\theta_{f_{dt}}$	✓	✓	✓	✓	✓	✓	✓
θ_{pt}	✓	✓	✓	✓	✓	✓	✓
Observations	21,519,511	21,519,511	21,519,511	21,519,511	21,519,511	21,519,511	21,519,511
Clusters (d)	225	225	225	225	225	225	225
Adj. R^2	0.744	0.744	0.744	0.744	0.744	0.744	0.744

Notes: Full firm-level panel (excluding Hong Kong and Macao), restricted to observations with valid sub-index data. Each column estimates a separate model replacing the aggregate EP index with one sub-index; the non-environmental depth control ($TD_{dt} \times g_p$ and $TD_{dt} \times b_p$) is included in all specifications but not shown. Fixed effects: firm-product-destination ($\theta_{f_{pd}}$), firm-destination-year ($\theta_{f_{dt}}$), product-year (θ_{pt}). Standard errors clustered at the destination level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

WB sub-indices: (1) Green Liberalization; (2) Enforcement & DSM. TREND sub-indices: (3) Green Market Access; (4) Enforcement & DSM; (5) Hard obligations; (6) Soft obligations; (7) Regulatory Space.

1.5 Dynamics, influence, and the extensive margin

Three final checks, each aimed at a different way the preceding results could be wrong.

The first is timing. Figure 1 plots the green and dirty gap relative to neutral products around entry into force, with never-treated destinations as controls. The pre-treatment coefficients are flat on both margins, which is the standard falsification of the identifying assumption. It should be read as exactly that and no more: parallel trends is a statement about the counterfactual path after treatment, and a flat pre-trend rules out the most detectable violation without establishing that the assumption holds thereafter. A mild negative drift appears for green products in the late post-period, but it is identified only by the four earliest cohorts and it disappears under the SUN2021 interaction-weighted estimator, whose aggregated treatment effect is -0.042 ($p = 0.27$) on the green gap and $+0.073$ ($p = 0.28$) on the dirty gap. The full event-study coefficients, the Sun–Abraham estimates, and the destination-specific trend specifications that address selection on trends are reported in the appendix.

The second is influence. A coefficient estimated from 23 treated destinations can be the product of one of them, and Table 8 re-estimates the dirty interaction dropping each in turn. The familiar concern does not materialise: the point estimate stays between -0.0097 and -0.0133 in all 23 samples, and dropping large partners such as India or Pakistan moves it by one to four per cent. What is fragile is not the estimate but its precision. Dropping Australia moves the coefficient by 13%, to -0.0103 , while nearly tripling its standard error from 0.0030 to 0.0087; dropping Korea doubles the standard error and gives -0.0097 . Australia and Korea are not outliers pulling the estimate — they are supplying the variation that identifies it, and without them the design cannot determine much of anything. Which of the two is pivotal, moreover, depends on the depth control: under DESTA it is Korea rather than Australia. A coefficient whose decisive observation is selected by a modelling choice unrelated to environmental content is behaving the way the permutation test already suggested it was.

The third is the margin itself. Everything above conditions on positive flows, so an agreement that created green exports to a market previously served by none would leave no trace in it. Table 9 estimates the same interactions by Poisson pseudo-maximum likelihood on an HS6–destination–year grid completed with zeros (SANTOSSILVA2006), of which 7.9 million cells of 8.2 survive the estimator’s removal of separated and singleton observations. The green interaction is $+0.0015$ ($p = 0.74$) with the World Bank index and $+0.0001$ ($p = 0.95$) with TREND: no trade creation. The dirty interaction is -0.030 ($p = 0.057$) and $+0.003$ ($p = 0.55$), marginal at best and, once again, only under one of the two codings. One limitation deserves stating: this grid detects a product newly exported to a destination, not a new firm beginning to export a product that other Chinese firms already ship there. The extensive margin is probed, not exhausted.

Table 8: Leave-one-out: EP×dirty coefficient stability across 23 treated destinations

Excluded destination	WB depth control		DESTA depth control	
	Excl. HK/Macao (1)	Incl. HK/Macao (2)	Excl. HK/Macao (3)	Incl. HK/Macao (4)
<i>None (baseline)</i>	−0.0119***	−0.0189***	−0.0113***	−0.0082*
<i>None, extended dirty</i>	−0.0121***	−0.0191***	−0.0107***	−0.0071
<i>High EP depth only</i>	−0.0271**	−0.0456***	−0.0161	−0.0164
Bangladesh	−0.0118***	−0.0188***	−0.0113***	−0.0082*
Brunei	−0.0119***	−0.0189***	−0.0114***	−0.0082*
Myanmar	−0.0118***	−0.0188***	−0.0112***	−0.0081*
Cambodia	−0.0119***	−0.0189***	−0.0112***	−0.0081*
India	−0.0120***	−0.0190***	−0.0115***	−0.0084**
Indonesia	−0.0119***	−0.0190***	−0.0119***	−0.0087**
Laos, PDR	−0.0119***	−0.0189***	−0.0113***	−0.0082*
Malaysia	−0.0124***	−0.0195***	−0.0110***	−0.0076*
Pakistan	−0.0114***	−0.0183***	−0.0106***	−0.0056
Philippines	−0.0118***	−0.0189***	−0.0115***	−0.0083*
Singapore	−0.0098**	−0.0166***	−0.0128***	−0.0100**
Korea Rep.	−0.0097*	−0.0225**	−0.0104	−0.0085
Sri Lanka	−0.0119***	−0.0189***	−0.0114***	−0.0082*
Thailand	−0.0120***	−0.0189***	−0.0115***	−0.0082*
Vietnam	−0.0122***	−0.0192***	−0.0112***	−0.0080*
Timor-Leste	−0.0119***	−0.0189***		
Iceland	−0.0119***	−0.0189***	−0.0112***	−0.0081*
Switzerland	−0.0133***	−0.0209***	−0.0115***	−0.0084*
Chile	−0.0121***	−0.0191***	−0.0104***	−0.0071
Costa Rica	−0.0114***	−0.0185***	−0.0109***	−0.0076*
Peru	−0.0133***	−0.0204***	−0.0121***	−0.0092**
Australia	−0.0103	−0.0245**	−0.0110***	−0.0048
New Zealand	−0.0119***	−0.0190***	−0.0113***	−0.0081*
Hong Kong		−0.0129***		−0.0112***
Macau		−0.0183***		−0.0082*
Observations	3,681,023	3,769,187	3,677,333	3,759,733
Clusters (<i>d</i>)	228	230	228	230

Notes: OLS on the collapsed panel. Dependent variable: log total exports from China. The coefficient shown is $\hat{\beta}$ on EP×dirty, re-estimated dropping one treated destination at a time. FE: $pd + dt + pt$. Standard errors (not shown) clustered by destination. Columns (1) and (3) exclude Hong Kong and Macao from the sample; columns (2) and (4) include them as additional treated observations. Columns (1)–(2) use WB total depth as the control-depth variable; columns (3)–(4) use DESTA depth. Observations and clusters refer to the baseline specification; individual rows differ slightly due to the dropped destination.

High EP depth only: estimated on the subsample excluding Peru, Switzerland, and Korea jointly — the three destinations with above-median WB EP depth.

Timor-Leste rows are blank in columns (3)–(4): Timor-Leste is absent from the DESTA dataset. Hong Kong and Macau rows are blank in columns (1) and (3): those territories are already excluded in those samples.

Note on inference. Stars are based on asymptotic p -values with destination-clustered SEs. With ≈ 23 treated destinations, these p -values are substantially understated relative to the wild cluster bootstrap: the baseline coefficient in column (1) has asymptotic $p < 0.001$ but bootstrap $p = 0.072$ (Table 1.5). Stars are not comparable to the rest of the paper.

How to read. This is a test of *coefficient stability*, not significance. The relevant question is how much the point estimate moves when one destination is dropped, not how many stars it carries.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ (asymptotic).

Table 9: PPML — extensive margin (collapsed panel, zero-filled)

	WB EP depth (EP_{dt}^{WB})		TREND EP count (EP_{dt}^{TR})	
	(1)	(2)	(3)	(4)
<i>Panel A — Non-environmental depth control (TD_{dt}, World Bank)</i>				
$EP_{dt} \times g_p$	0.0015 (0.0046)	0.0029 (0.0061)	0.0001 (0.0019)	−0.0001 (0.0021)
Asymptotic p	[0.738]	[0.630]	[0.951]	[0.972]
$EP_{dt} \times b_p$	−0.0301* (0.0158)	−0.0268 (0.0178)	0.0028 (0.0047)	0.0048 (0.0050)
Asymptotic p	[0.057]	[0.133]	[0.550]	[0.335]
θ_{pd}	✓	✓	✓	✓
θ_{dt}	✓	✓	✓	✓
θ_{pt}	✓	✓	✓	✓
Observations	7,895,543	8,030,137	7,895,543	8,030,137
Clusters (d)	225	228	225	228
<i>Panel B — DESTA depth control</i>				
$EP_{dt} \times g_p$	0.0053 (0.0045)	0.0064 (0.0050)	0.0010 (0.0013)	0.0008 (0.0015)
Asymptotic p	[0.232]	[0.205]	[0.442]	[0.565]
$EP_{dt} \times b_p$	−0.0186 (0.0169)	−0.0108 (0.0198)	0.0027 (0.0053)	0.0046 (0.0050)
Asymptotic p	[0.270]	[0.585]	[0.612]	[0.358]
θ_{pd}	✓	✓	✓	✓
θ_{dt}	✓	✓	✓	✓
θ_{pt}	✓	✓	✓	✓
Observations	7,878,784	8,003,772	7,878,784	8,003,772
Clusters (d)	225	228	225	228

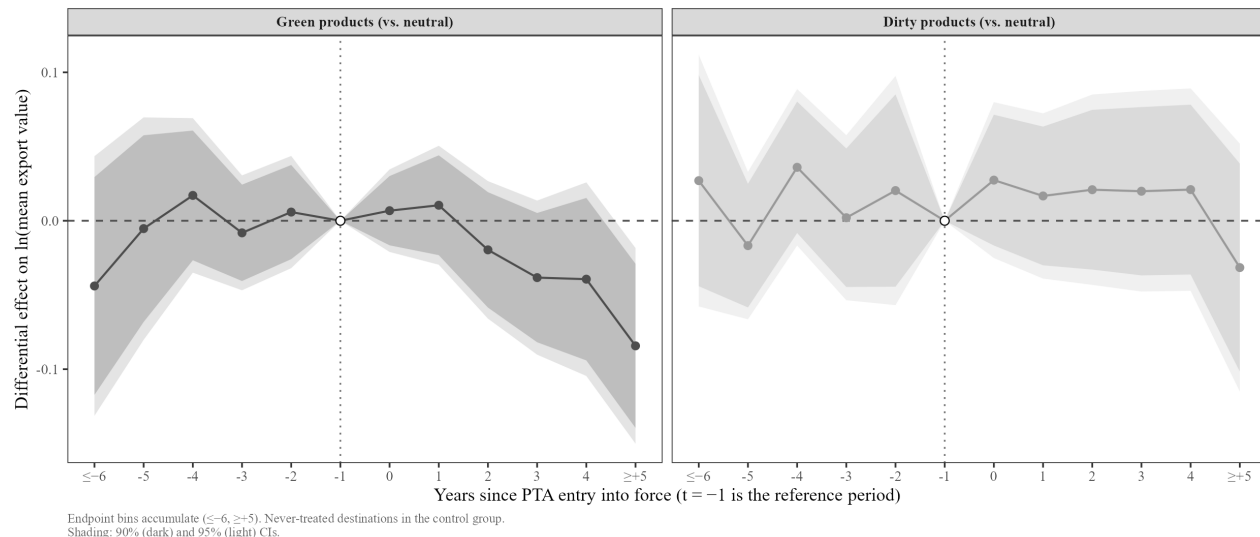
Notes: PPML estimated on the HS6–destination–year grid completed with zeros (Santos Silva & Tenreyro, 2006); fixed effects: product×destination (θ_{pd}), destination×year (θ_{dt}), product×year (θ_{pt}); standard errors clustered at the destination level. Columns (1) and (3) exclude Hong Kong and Macao; columns (2) and (4) include them. Panel A controls for non-environmental WB depth ($TD_{dt} \times g_p$ and $TD_{dt} \times b_p$, not shown); Panel B uses the DESTA depth index.

Log-linear estimates condition on positive flows: if an environmental clause created new green exports to a market where none previously existed, those estimates would miss it. PPML retains zeros and therefore captures this margin.

PPML coefficients are approximate percentage changes.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ (asymptotic).

Figure 1: Event study: green and dirty vs. neutral products around PTA entry into force



Notes: Collapsed panel; FE $pd + dt + pt$; destination-clustered 95% confidence bands; $t = -1$ is the reference period. Coefficients are tabulated in Appendix Table 1.5. Source: Chinese Customs Office; World Bank DTA Database.

Note for the author

In appendix, cited in the text above: full saturation ladder for all three outcomes (1.5, 1.5); Sun–Abraham event study (1.5); tabulated TWFE event study (1.5); destination-specific trends and pre-trend tests (1.5); Hong Kong and Macao variants of both main tables (1.5, 1.5); complete bootstrap and permutation tables (1.5, 1.5); the dirty-coefficient matrix (1.5); full-panel robustness battery (1.5); APEC versus OECD product definition (1.5); CO₂ intensity (1.5); collapsed-panel sub-indices (1.5); minimum detectable effects (1.5) and the Brandi benchmark (1.5). C-overlap is dropped, as agreed.

Three things to decide. (i) The bounding argument — minimum detectable effects and the comparison with BRANDI2020 — has no home in this version. It is what turns “we find no effect” into “we can rule out effects above 3% per provision”. It would fit as two paragraphs at the end of Section 1.1 or as the opening of the conclusion. (ii) The wild cluster bootstrap was never run on the full-panel sub-indices; Section 1.4 currently reports asymptotic p -values and says so. (iii) The compact ladder in Table 2 is new; if you would rather keep the section leaner it moves to the appendix and the argument survives as one sentence.

Appendix placeholders (labels only)

- Saturation ladder
- Sun–Abraham
- Event study TWFE
- Destination trends
- Full panel incl. HK/Macao
- Collapsed incl. HK/Macao
- Bootstrap
- Permutation
- Dirty matrix
- Robustness battery
- APEC vs OECD
- CO₂ intensity
- Collapsed sub-indices
- Minimum detectable effects
- Brandi benchmark